

Bimanual Garment Folding via Generative Action Chunking: A Comparative Study from Diffusion Policy to ACT with Geometric Priors

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Abstract

Robotic manipulation of highly deformable garments presents significant challenges due to their high dimensionality and complex dynamics. This paper presents an exploratory study of diverse imitation learning strategies within the context of the ICRA 2026 LeHome Challenge. We investigate a progression of architectures and state-of-the-art visuomotor policies, evolving from Diffusion Policy through a sequence of Vision-Language-Action (VLA) models—including SmolVLA, $\pi_{0.5}$, and X-VLA—ultimately converging on Action Chunking with Transformers (ACT). Our study systematically evaluates the impact of mid-level geometric priors (colorized depth) provided as an additional input alongside RGB, robust data augmentations, and Parameter-Efficient Fine-Tuning (PEFT) using LoRA. We further analyze the effects of critical hyperparameters such as action chunk size and execution steps. Our results demonstrate that while VLA-based models show strong semantic understanding, the fine-grained bimanual coordination required for garment folding is best captured by ACT with geometric and temporal optimizations. We report per-garment success rates, showing significant performance gains across categories, with a final macro-average success rate of 61.25% and enhanced generalization to unseen garment styles.

1. Introduction

Robotic garment manipulation is a benchmark for dexterous bimanual control. Unlike rigid object manipulation, handling textiles requires reasoning about infinite-dimensional configuration spaces and managing complex self-occlusions. The ICRA 2026 LeHome Challenge [16] provides a rigorous platform for evaluating these capabilities, requiring robots to fold long-sleeved tops, short-sleeved tops, long pants, and shorts in a high-fidelity simulation environment [14].

This paper documents our systematic exploration of various visuomotor policies and visual representations to solve the bimanual folding task. Rather than focusing on a sin-

gle architecture, we treat this project as an empirical study of the most promising strategies in modern imitation learning. Our journey followed a deliberate progression: beginning with Diffusion Policy [6], we then explored the emerging class of Vision-Language-Action (VLA) models, including SmolVLA [5], $\pi_{0.5}$ [11], and X-VLA [1]. Ultimately, we transitioned to Action Chunking with Transformers (ACT) [19] to better capture the high-frequency coordination required for bimanual manipulation.

Furthermore, we evaluate the role of **geometric priors** by providing colorized depth maps as an additional input alongside RGB, the impact of **data augmentations** on cross-style generalization, and the utility of **Parameter-Efficient Fine-Tuning (PEFT)** via LoRA [10] for adapting large-scale foundation models. Furthermore, we perform a sensitivity analysis on temporal parameters, specifically the action chunk size and the number of execution steps.

Our results provide a comprehensive view of the strengths and weaknesses of current architectures. While diffusion-based and VLA-based models demonstrate impressive flexibility, we find that the transformer-based temporal modeling in ACT, when combined with geometric priors and robust augmentations, provides the most reliable performance for the specific constraints of the LeHome Challenge. We report detailed results across all garment categories, providing insights into the specific failure modes and generalization capabilities of each strategy.

In summary, our contributions include:

- A comparative study of multiple visuomotor architectures (DP, SmolVLA, $\pi_{0.5}$, X-VLA, ACT) for bimanual garment folding.
- An evaluation of mid-level visual representations (colorized depth) and their impact on reducing appearance-based overfitting.
- A systematic analysis of data augmentation and PEFT strategies for improving generalization to unseen garment styles.
- A detailed breakdown of success rates per garment category and an analysis of critical temporal hyperparameters.

2. Related Work

2.1. Robotic Garment Manipulation

Manipulating highly deformable textiles presents unique challenges compared to rigid objects due to infinite-dimensional configuration spaces and complex dynamics [8, 13]. Recent frameworks have automated tasks like smoothing and folding by employing dense visual correspondences, learned keypoints, and deep imitation learning [7, 9]. Systems like SpeedFolding [2] and UniFolding [17] leverage bimanual coordination for efficient folding, often using discrete primitives or canonical alignments. Our work explores the potential of continuous action spaces modeled by generative architectures.

2.2. Visuomotor Imitation Learning

Imitation learning has shifted from traditional behavioral cloning to generative models that handle multi-modality. Diffusion Policy [6] models the action distribution via a diffusion process, while Action Chunking with Transformers (ACT) [19] leverages the temporal modeling of Transformers to predict chunks of actions, mitigating compounding errors. Recent advancements have introduced Large Vision-Language-Action (VLA) models such as RT-2 [4], OpenVLA [12], and the π series (π_0 , $\pi_{0.5}$) [3, 11]. These models often use text tokens to encode actions and leverage web-scale knowledge for robotic control. We investigate how these foundation models adapt to the specialized domain of garment folding.

2.3. Generalization and Adaptation

Achieving robust generalization across object styles requires effective visual representations and efficient adaptation techniques. 3D Diffusion Policy (DP3) [18] uses point clouds to overcome appearance bias. For adapting large-scale models, Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA [10] allow for rapid task-specific optimization without re-training the entire backbone. Furthermore, online adaptation frameworks like DAgger [15] provide a mechanism for human-in-the-loop (HIL) correction to address failure cases, a paradigm we explore within the constraints of our training infrastructure.

3. Method

Our methodology followed an iterative exploration of models and representations, progressing from standard diffusion-based policies to large-scale Vision-Language-Action (VLA) models, and finally to optimized action chunking architectures. This section details each stage of our exploration and the specific enhancements implemented.

3.1. Phase 1: Initial Baseline (Diffusion Policy)

We began our experiments with Diffusion Policy (DP) [6], which has demonstrated strong performance in diverse manipulation tasks by modeling action trajectories as a denoising process. However, we found that DP struggled with the high-precision coordination required for bimanual garment folding, often producing inconsistent joint targets.

3.2. Phase 2: Vision-Language-Action Models (SmolVLA, $\pi_{0.5}$, and X-VLA)

We then explored the emerging class of Vision-Language-Action (VLA) models. We first evaluated SmolVLA [5], a compact VLA model that integrates visual and language tokens. SmolVLA provided better semantic understanding of the task instructions but lacked the fine-grained control necessary for handling deformable fabrics. To leverage more powerful foundational knowledge, we then explored the $\pi_{0.5}$ model [11], an evolution of π_0 that uses a combination of discrete tokens and continuous flow matching for dexterous control. We also evaluated X-VLA [1], which focuses on cross-modal alignment specifically for complex manipulation. While these models showed improved robustness in identifying garment parts (sleeves, hems), their inference latency on our hardware and the challenges of adapting their large backbones to the specific dynamics of IsaacLab led us to seek more efficient alternatives.

3.3. Phase 3: Action Chunking with Transformers (ACT)

Our final and most successful architecture was Action Chunking with Transformers (ACT) [19]. ACT addresses the "compounding error" problem by predicting chunks of actions simultaneously using a CVAE and a Transformer decoder. This approach is particularly well-suited for bimanual tasks as it inherently captures the temporal correlations between the two arms.

3.4. Geometric and Generalization Enhancements

Across all models, we investigated several strategies to improve robustness and generalization:

- **Colorized Depth:** To provide richer geometric information, we augmented the raw RGB observations with depth maps colorized using a JET-style gradient. This allows the model to leverage pre-trained visual features while explicitly perceiving the 3D structure of the cloth, such as folds and wrinkles.
- **Data Augmentations:** We implemented a suite of augmentations, including random cropping, rotation, and color jittering (on the colorized depth), to simulate variations in garment pose and sensor noise.
- **PEFT via LoRA:** For the VLA models (SmolVLA, $\pi_{0.5}$), we employed Parameter-Efficient Fine-Tuning (PEFT) using Low-Rank Adaptation (LoRA) [10]. This allowed

us to adapt the massive pre-trained backbones to the garment folding domain by training only a small subset of parameters, preventing catastrophic forgetting while improving task alignment.

- **Hyperparameter Tuning:** We performed a systematic analysis of the ACT temporal parameters. Specifically, we varied the `chunk_size` (number of actions predicted in one chunk) and the `n_action_steps` (number of execution steps before the next chunk is predicted) to find the optimal balance between long-horizon planning and real-time responsiveness.

3.5. Human-in-the-Loop and Infrastructure

We also investigated the feasibility of Human-in-the-Loop (HIL) corrections using a DAgger-style framework [15]. The intent was to capture failure cases and provide corrective demonstrations. However, as noted in Section 4.4, the lack of a graphical interface on our HPC environment made real-time human intervention impractical.

4. Experiments and Results

4.1. Quantitative Analysis

We evaluate our models on the four garment categories of the LeHome Challenge: long-sleeved tops (T-L), short-sleeved tops (T-S), long pants (P-L), and shorts (P-S). The evaluation uses the success rate metric defined by the alignment of final cloth vertices against a canonical folded state.

Table 1 presents the success rates for each model across all garment categories. This per-garment breakdown reveals that while certain models (like SmoVLA) excel in specific categories (e.g., shorts), they struggle with more complex geometries like short-sleeved tops.

The transition from DP to VLA models and finally to ACT shows a clear trend of improvement. ACT architectures, particularly when augmented with depth and geometric priors, show a more balanced performance across all garment types. Notably, the integration of depth as an additional input significantly improved success in the challenging "Top Short" and "Pant Long" categories by providing better spatial disambiguation.

4.2. Generalization to Unseen Styles

A critical component of our study is the model’s ability to generalize to unseen garment styles. Table 2 shows the success rates on held-out garments per category.

While ACT + Depth achieves the highest macro-average on seen garments, the addition of data augmentations (ACT + Depth + Aug) results in the best generalization performance, particularly for the pant categories. This indicates that augmentations effectively regularize the model, preventing it from overfitting to the specific training geome-

tries and improving zero-shot transfer to novel garment instances.

4.3. Sensitivity Analysis: Chunk Size and Action Steps

We investigated the impact of the action chunk size and the number of execution steps in ACT. Our findings suggest that the `chunk_size` is a critical hyperparameter for long-horizon bimanual tasks.

- **Small Chunks (≤ 30):** The model exhibits disjointed movements and fails to capture the smooth coordination needed for the full folding sequence.
- **Medium Chunks (50-100):** Provides the best balance, allowing the model to plan a coherent trajectory while remaining responsive to the current state.
- **Large Chunks (≥ 150):** Leads to an accumulation of errors; if the initial prediction is slightly inaccurate, the robot drifts significantly from the desired path before the next chunk is predicted.

The number of `n_action_steps` (the execution window) also plays a role in trajectory smoothness. We found that execution steps slightly smaller than the chunk size (e.g., 50 steps for a chunk of 60) allow for temporal smoothing during the transition between chunks.

4.4. Failure Analysis and Limitations

Failure modes vary across models. DP often results in catastrophic entanglement, while VLA models tend to misidentify the specific part of the cloth to grasp. ACT failures are typically "graceful," such as a missed grasp that leaves the garment in a stable state. The "Top Short" category remains the most difficult due to the minimal surface area for grasping the sleeves. Infrastructure constraints, such as the lack of a GUI for real-time human intervention, limited our ability to address these edge cases through online corrective feedback.

5. Conclusion

In this work, we explored the application of generative imitation learning for bimanual garment folding in the context of the ICRA 2026 LeHome Challenge. Our investigation led us to shift from our initial 3D Diffusion Policy (DP3) proposal to an architecture based on Action Chunking with Transformers (ACT). We demonstrated that ACT’s ability to model action sequences as chunks is significantly better suited for the coordinated bimanual trajectories required in cloth manipulation compared to standard Diffusion Policy.

By integrating geometric priors through colorized depth and employing extensive data augmentations, we achieved a macro-average success rate of 61.25% and demonstrated robust generalization to unseen garment styles. Our findings highlight the importance of choosing architectures that

Table 1. Per-garment success rates (%) across different architectures and strategies. T-L: Top Long, T-S: Top Short, P-L: Pant Long, P-S: Pant Short. Macro-Avg is the mean across all categories. Bold values indicate the best performance in each category.

Model Strategy	T-L (%)	T-S (%)	P-L (%)	P-S (%)	Macro-Avg (%)
Diffusion Policy (DP) [6]	5.00%	0.00%	3.00%	38.33%	11.67%
SmolVLA [5]	56.67%	11.67%	35.00%	73.33%	44.17%
SmolVLA + Augmentations	55.00%	20.00%	30.00%	81.67%	46.67%
SmolVLA + Aug + PEFT (LoRA)	65.00%	28.33%	43.00%	78.33%	53.75%
ACT (Base) [19]	75.00%	45.00%	38.00%	78.33%	58.96%
ACT + Depth	70.83%	50.00%	47.00%	77.50%	61.25%
ACT + Depth + Aug	61.67%	43.33%	50.00%	84.17%	59.79%

Table 2. Per-garment success rates (%) on unseen garment styles.

Model	T-L (%)	T-S (%)	P-L (%)	P-S (%)	Unseen Avg (%)
SmolVLA [5]	30.00%	0.00%	20.00%	60.00%	27.50%
ACT (Base)	85.00%	30.00%	30.00%	45.00%	47.50%
ACT + Depth	60.00%	35.00%	10.00%	55.00%	40.00%
ACT + Depth + Aug	45.00%	35.00%	70.00%	55.00%	51.25%

can capture complex temporal dependencies and the effectiveness of mid-level geometric representations in reducing appearance-based overfitting.

Future work includes resolving the infrastructure limitations to implement human-in-the-loop correction and further exploring the potential of Large Vision-Language-Action models through more extensive parameter-efficient fine-tuning. We believe that combining the strengths of transformer-based action chunking with the vast pre-trained knowledge of VLAs is a promising direction for achieving more generalizable and capable robotic garment manipulation systems.

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